Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **finwing**
b. Team leader name: **Hrutik Kumthekar**
c. Problem Statement: **Media, Content & Digital Experiences** — turning static learning content into playable, verified game worlds

BRIEF ABOUT THE IDEA

Upload any chapter. Play it. Win only by using the concept

Upload your chapter
Notes, a textbook chapter or a PDF. Gemini pulls out the concepts, the rules and the usual misconceptions.

Gemini builds a world
A story RPG with a villain, a twist and one concept per act. Nano Banana paints it.

Win by using the idea
Drive, aim, order a process, pin evidence, tune a RAG pipeline. No multiple choice.



Seven Sharp at the Haveli: built by Gemini from a RAG textbook chapter, played with live retrieval over that same chapter.

Gemini writes it, Nano Banana paints it — and the game's own code proves every level **winnable, unguessable and grounded in your notes** before you play.

OPPORTUNITIES

Others skin a quiz. We make the concept the controller

	Prompt-to-game tools	Notes-to-quiz apps	Quiz RPG makers	Chapter Quest
Reads your own material	✗	✓	~	✓
You win by using the concept, not answering MCQs	~	✗	✗	✓
Every level proved winnable and grounded in your notes	✗	✗	✗	✓
Adapts to the misconception you just showed	✗	~	✗	✓

✓ yes ~ partly ✗ no — categories as we assessed them, not an endorsement by any vendor

How it solves the problem
Reading is passive. Here you predict, act, and see why — and a character who embodies your exact mistake walks in to argue it.

USP: verified before you play
Every act is proved winnable; guessing wins ≤6%; every rule is quoted word for word from your notes; Gemini play-tests its own levels.

The opportunity
Every learner already has the content. We turn it into an interactive digital experience in about two minutes.

LIST OF FEATURES OFFERED BY THE SOLUTION

Four ways to play, every level verified before you play

Formula worlds
Drive a train, aim a launch, balance a beam. A real simulator decides, never the AI.

Sequence boards
Order a process or a causal chain. A broken link explains why, quoted from your notes.

Evidence boards
Pin each document to the claim it proves. Decoys explain what they really show.

Live RAG worlds
Tune chunking, top-k and grounding over your own chapter, with Gemini embeddings.

The ✓ verified card
See what was proved: start loses, a win exists, guess odds, every citation.

Misconception characters
Miss, and the character who embodies your wrong idea walks in. Reply in ~1.5 s.

Predict, then compare
Commit to a guess first; after the win, see it against reality — and where the analogy breaks.

Painted worlds
Nano Banana paints the cast, props and rooms in one consistent style.

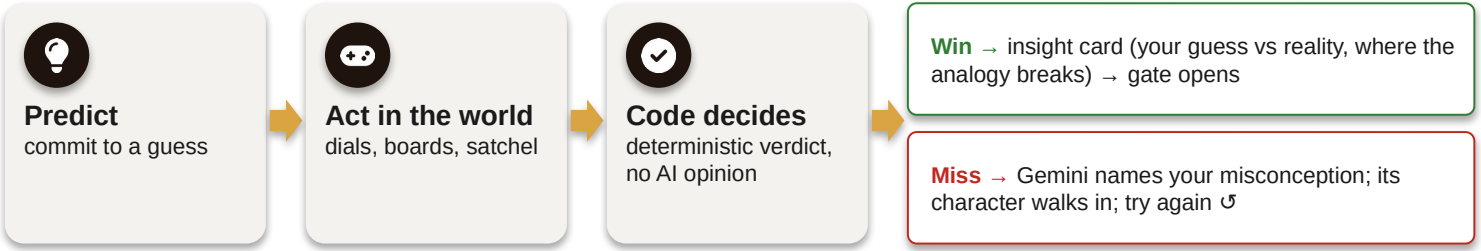
PROCESS FLOW DIAGRAM OR USE-CASE DIAGRAM

Chapter to verified game in ~2 minutes, then it adapts to you

BUILD once per chapter, ~2 min, cached forever



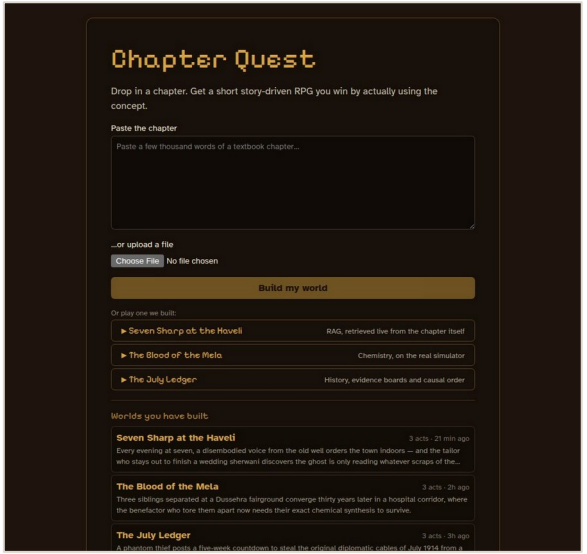
PLAY every attempt

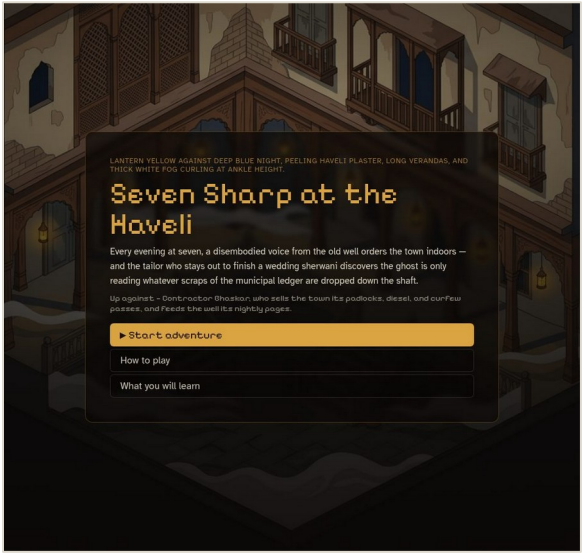


Every verdict comes from the game's code; Gemini explains, never judges a win.

WIREFRAMES/MOCK DIAGRAMS OF THE PROPOSED SOLUTION (OPTIONAL)

Not mock-ups: real screens from the running prototype







1. Paste or pick
Drop in a chapter, or open a world we built: RAG, chemistry, history.

2. A world, not a worksheet
Every chapter gets a title, a cast, a villain and a painted style.

3. Predict first
A guess before the act, compared with reality after it.

ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION

One Cloud Run service, Gemini at every stage, code as referee



Deploy: Docker → Cloud Build / Artifact Registry → Cloud Run (asia-southeast1) · Gemini via Vertex AI with the service account, no API keys · built worlds ship in the image, so the demo opens instantly

TECHNOLOGIES TO BE USED IN THE SOLUTION

Gemini on Vertex AI, used five ways — Google Cloud end to end

Model	What it does here	Gemini capability
Gemini 4 Argon	Curriculum, story drafts and judge, act writing, evidence judge	Structured output (JSON schemas), PDF input
Gemini 3.8 Flash	Play-tester agent, Oracle answers, RAG judge	Function calling (tool use)
Gemini 3.5 Flash-Lite	Guide, misconception diagnosis, reranking	Low-latency structured output
Nano Banana 2.1	Cast, props, rooms, walk sheets	Image generation with a style reference
Gemini Embedding	Live retrieval over the uploaded chapter	Embeddings

Cloud & app stack

Vertex AI · Cloud Run · Cloud Build · Artifact Registry

TypeScript monorepo

React 18 + Vite · Phaser 3

Node 20 + Express · Docker

node:test — 231 tests

Playwright headless QA · pdf.js

Everything the AI makes is structured data. Nothing it writes is executed — the game's own code runs every check.

ESTIMATED IMPLEMENTATION COST (OPTIONAL)

Pay once per chapter. Every replay is free

	What it takes	Basis
Build a world (script)	≈13 model calls, ≈95k tokens	Measured: average of 8 chapters
Paint a world	16–21 Nano Banana images	Measured; 3 painted at once, ~2 min
Replay a built world	$0 — scripts, art and RAG runs cached	Measured: cache keyed by chapter hash
One adaptation (per miss)	1 Flash-Lite call, ~1.5 s	Measured latency
Hosting	One Cloud Run container, scales to zero	745 MB image

≈$0.20–0.40

to write and verify one world's script*

$0

to replay it, for any number of learners

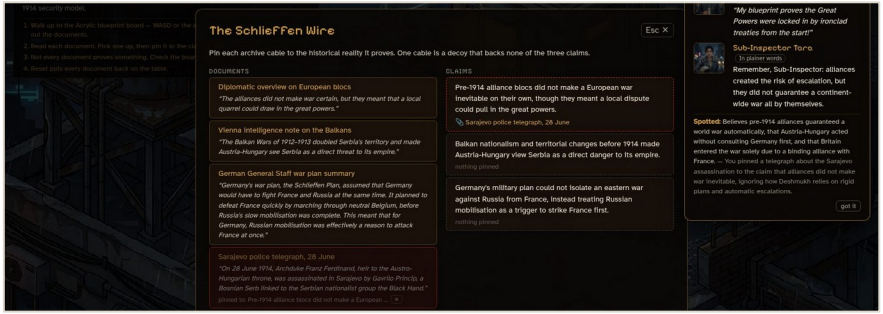
~1.5 s

for a misconception-aware hint

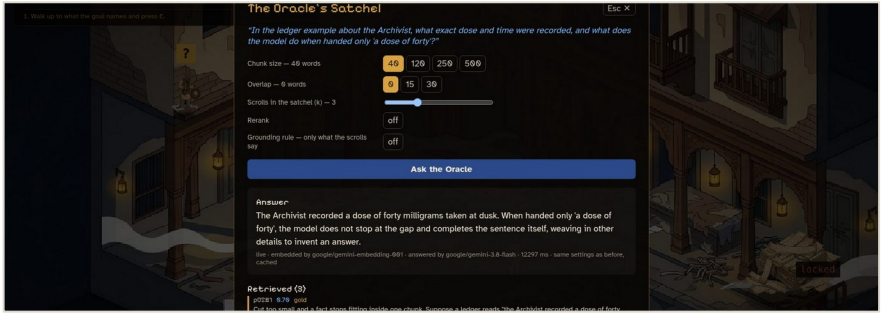
*Estimate at the per-token prices our router reports for Gemini 4 Argon ($2 in / $10 out per 1M tokens); the range covers the unmeasured input/output split. Image pricing excluded.

SNAPSHOTS OF THE PROTOTYPE

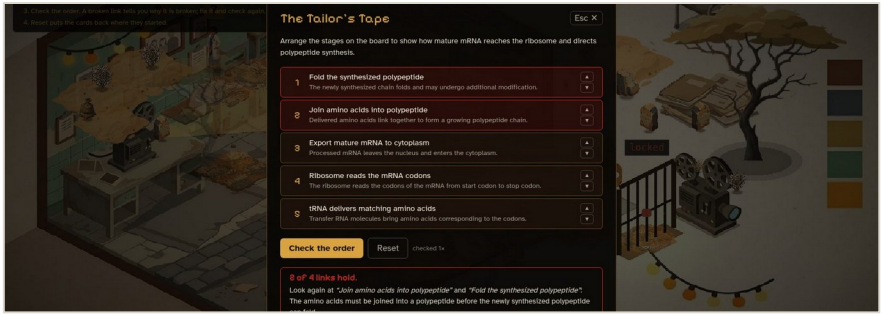
Snapshots from the working prototype



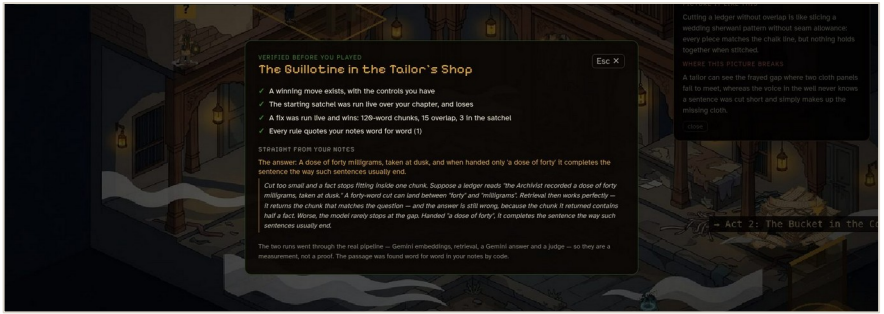
Pin the wrong document — the character who embodies your mistake walks in



Live RAG over the chapter: a right answer still fails — the fact was torn



Order a process; a broken link explains itself, quoted from your notes



The ✓ verified card: what was proved before you played

PROTOTYPE PERFORMANCE REPORT/BENCHMARKING

Measured, not claimed: 8 real chapters rebuilt from scratch

8/8
chapters became a game from their own content

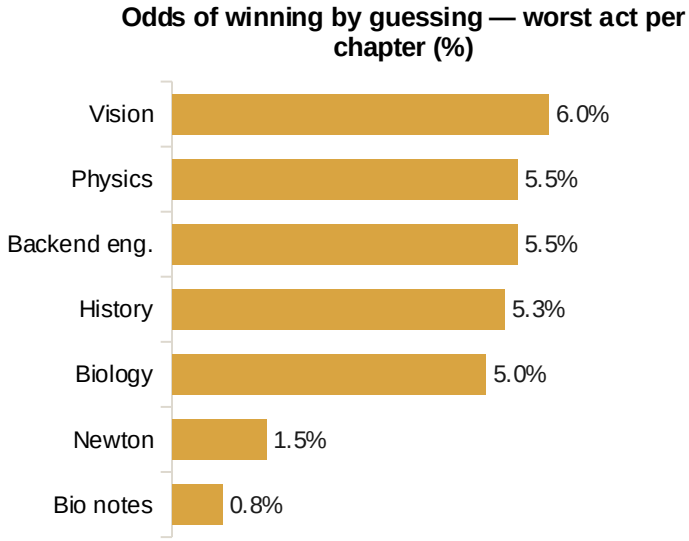
22/22
acts proved finishable

≤6%
worst-case odds of winning by guessing

74
citations checked word for word against notes

3/3
RAG acts run live: start loses, a fix wins

2.5×
faster art: 319 s → 128 s for 16 images



Our bar for every act: random play must almost always lose. Build: ~2 min per world, 13 calls. Eval run 18 Oct 2026 with live Gemini; the evidence judge is a model and is labelled as one.

ADDITIONAL DETAILS/FUTURE DEVELOPMENT (IF ANY)

Next: every kind of concept, every classroom, every language

NEXT 4 WEEKS

Definitions & arguments
a mechanic for the last 8% of concepts we refuse today

Gemini vision art QA
every painted asset checked against what it should show

Context caching
the chapter cached once per build: faster, cheaper

NEXT QUARTER

Teacher dashboard
class-wide misconception maps from real attempts

Indian languages
worlds written and voiced in the learner's language

Classroom & LMS
assign a chapter, track who has played it

VISION

Voice characters
argue with the misconception, out loud (Gemini Live)

Publisher toolkit
any content library → verified interactive worlds

Open verifier
the checks as a standard for generated learning games

Provide links to your:

- GitHub Public Repository: **https://github.com/capsemble/akagami**
- Demo Video Link (3 Minutes): **[ADD: 3-minute demo video link]**
- Final Product Link: **https://chapter-quest-140843079737.asia-southeast1.run.app**

Try it in 60 seconds

1. Open the product link and pick “Seven Sharp at the Haveli” — a RAG chapter turned into a game.
2. Ask the Oracle with the torn 40-word chunks: the answer is right, the act still fails. Fix the chunking — it passes.
3. Open “The July Ledger”, pin the wrong document, and watch the misconception’s own character walk in.
4. Or paste your own notes and watch the build verify every level live.

HOW CHAPTER QUEST MEETS THE JUDGING CRITERIA

Why finwing's Chapter Quest deserves to win

40%

Technical merit & Gen AI

— Gemini at every stage: 5 models, structured output, tools, images, embeddings

— Code is the referee: acts proved winnable, 231 tests, 8-chapter eval

— Agentic: Gemini play-tests and repairs its own levels

22/22

acts proved finishable

25%

Impact

— Any learner's own notes become a game in ~2 minutes

— Learning by doing: predict, act, see why

— 18+ learners, exam prep, upskilling in applied AI

8/8

chapters became games

25%

Innovation

— Verified generative games, not prompt-and-hope

— Misconception characters driven by your real attempts

— RAG taught with live RAG over the chapter itself

3/3

RAG acts verified live

10%

User experience

— Paste and play — no setup

— Every verdict explains itself, quoted from your notes

— Painted worlds; drag or click to play boards

~1.5 s

to a tailored hint

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Thank you

